

INRE Project — Modelling Methodology

Document type: Methodology (input data, sources, mathematical formulation, data processing)
Model: PyPSA-Eur with INRE extension layer
Geographic scope: Germany (DE), electricity-only
Planning horizon: 2050 technology costs, 2024 renewable fleet, January 2021 weather window

1. Purpose and scope

This document describes the methodology used in the **INRE** (Integrated Nuclear & Renewable Energy) study. The model simulates the German electricity system under normal winter conditions and under **Dunkelflaute** stress — extended periods of simultaneously low wind and solar output — with optional investment in advanced nuclear technologies (SMR, MSR, LFR).

The workflow builds on [PyPSA-Eur](#), an open-source European energy system model, and adds a custom pre-solve layer that:

- 1. Derates variable renewable energy (VRE) availability profiles for Dunkelflaute scenarios.
- 2. Adds extendable advanced nuclear generators at candidate sites with technology-specific costs and operational parameters.

In scope: transmission-constrained dispatch, capacity expansion, storage, CO₂ cap, 2-week winter snapshot optimisation.
Out of scope: sector coupling (heat, transport, industry), cross-border trade, unit commitment, existing German nuclear (all shut down).

2. Modelling framework

2.1 Software stack

Component	Role
PyPSA	Linear optimal power flow and capacity expansion
PyPSA-Eur	European grid build, clustering, data retrieval
Snakemake	Reproducible workflow orchestration
Atlite	Weather-to-renewable profile conversion
HiGHS	LP solver (free, no commercial licence required)

2.2 Workflow

```
retrieve data → build base network → cluster (10 nodes) → renewable profiles
  → add_electricity → prepare_network
    → [INRE: apply_inre_network] ← Dunkelflaute and/or nuclear modifications
      → solve_network (LP)
        → results and comparison
```

The INRE step ([scripts/inre/apply_inre_network.py](#)) runs **after** [prepare_network](#) and **before** [solve_network](#) when Dunkelflaute stress and/or new nuclear carriers are enabled. The base scenario skips this step

and passes the prepared network directly to the solver.

2.3 Spatial and temporal resolution

Setting	Value	Rationale
Country	Germany only (DE)	Focused national adequacy study
Spatial clusters	10 nodes	Balance between north–south heterogeneity and runtime
Snapshot window	2021-01-25 → 2021-02-08	Documented Jan 2021 winter Dunkelflaute episode
Temporal resolution	3 hours (112 snapshots)	Target ≤ 15 min solve per scenario
Weather cutout	europe-2021-sarah3-era5	ERA5 + SARA3-3 for 2021
Cost year	2050	PyPSA technology-data planning horizon
Renewable fleet year	2024	powerplantmatching capacity estimate

3. Input data and sources

3.1 Summary table

Data category	Primary source	Access	INRE override
Transmission grid topology	OpenStreetMap (via PyPSA archive)	Free (data.pypsa.org)	No
Power plant database	powerplantmatching	Free	Filter for DE, retirement dates
Renewable installed capacities (2024)	powerplantmatching estimate	Free	No
Electricity demand	ENTSO-E Transparency Platform	Free	No
Weather (wind, irradiance, temperature)	ERA5 + SARA3-3 via Atlite	Free (data.pypsa.org)	Dunkelflaute derating applied in stress scenarios
Technology costs (2050)	PyPSA technology-data	Free	Nuclear carriers overridden
Advanced nuclear techno-economics	OECD/NEA, IEA + INRE assumptions	Literature	Yes — data/inre/custom_costs_nuclear.csv
Dunkelflaute severity	INRE scenario parameters	Manual	Yes — data/inre/dunkelflaute.yaml

Data category	Primary source	Access	INRE override
Nuclear candidate sites	Former German NPP locations	Manual	Yes — <code>data/inre/custom_powerplants_nuclear_DE.csv</code>

3.2 Grid and topology

The base network is retrieved from the PyPSA-Eur archive (OpenStreetMap-derived transmission lines and substations). Germany is isolated (`countries: [DE]`) and clustered to **10 zones** using PyPSA-Eur's `cluster_network` rule. Each zone aggregates:

- Nodal electricity demand (ENTSO-E, distributed to buses),
- Conventional generators from powerplantmatching,
- Renewable generators with Atlite-derived availability profiles,
- Inter-zonal AC transmission lines (22 lines after clustering).

3.3 Conventional and renewable fleet

Power plant filter (config):

(DateOut > 2025 or DateOut != DateOut) and (DateIn < 2026 or DateIn != DateIn)

This retains plants without a recorded shutdown date or with shutdown after 2025, and excludes plants commissioned from 2026 onward. Existing German nuclear is **not** included (all commercial plants shut down by April 2023).

Renewable capacities are estimated from powerplantmatching for year **2024** with technology mapping:

powerplantmatching	PyPSA carrier
Onshore	<code>onwind</code>
Offshore	<code>offwind-ac</code>
PV	<code>solar</code>

Extendable technologies (base case): solar, solar-hsat, onwind, offwind-ac, offwind-dc, offwind-float, OCGT, CCGT, battery, H₂ store, H₂ pipeline.

3.4 Weather and renewable profiles

Weather data are taken from the `europe-2021-sarah3-era5` cutout:

- **ERA5**: wind speed, temperature, and related meteorological fields.
- **SARAH-3**: satellite-derived surface solar irradiance.

Atlite converts weather to per-generator availability time series `p_max_pu` (per-unit maximum output). Offshore depth constraints are disabled (`max_depth: false`) to allow unrestricted offshore siting in the model.

3.5 Technology costs

Default costs come from PyPSA technology-data for planning year **2050**, including:

- Overnight investment (EUR/kW),
- Fixed O&M (FOM, %/year),

- Variable O&M (VOM, EUR/MWh),
- Fuel costs,
- Electrical efficiency,
- Component lifetime,
- CO₂ intensity.

Nuclear new-build costs are **overridden** via `costs.custom_cost_fn: data/inre/custom_costs_nuclear.csv`, which is merged into the standard cost database during the build phase.

3.6 Advanced nuclear technology parameters

Techno-economic parameters for SMR, MSR, and LFR are documented in `nuclear-reactor-datasheet.md` and implemented in `data/inre/custom_costs_nuclear.csv`. Investment costs include **+15% decommissioning** on top of overnight CAPEX.

Parameter	SMR	MSR	LFR	Unit	Source
CAPEX (overnight)	5,000	5,800	5,400	EUR/kW	OECD/NEA, IEA
Decommissioning	+15%	+15%	+15%	of CAPEX	OECD/NEA, IEA
Total investment	5,750	6,670	6,210	EUR/kW	Computed
Fixed O&M	40	45	42	EUR/kW-year	OECD/NEA, IEA
Variable O&M	3.0	3.5	3.2	EUR/MWh	OECD/NEA, IEA
Fuel cost	3.0	2.5	2.8	EUR/MWh _{th}	INRE uranium estimate
Electrical efficiency	0.33	0.35	0.34	p.u.	INRE assumption
Lifetime	60	50	55	years	INRE assumption
CO ₂ intensity	0	0	0	t/MWh	INRE assumption
Ramp rate limit	0.50	0.50	0.50	p.u./hour	INRE assumption
Site capacity limit	1,500	1,500	1,500	MW	Config default

Operational parameters applied in `scripts/inre/add_nuclear_technologies.py`:

Parameter	Value	Description
<code>p_max_pu</code>	0.90	Availability factor (90%)
<code>p_min_pu</code>	0.30	Minimum stable load (30% of rated capacity)
<code>ramp_limit_up/down</code>	0.50	Maximum ramp rate (per unit per hour)
<code>p_nom_max</code>	1,500 MW	Maximum build per candidate site

3.7 Nuclear candidate sites

Candidate locations are former German nuclear power plant sites in `data/inre/custom_powerplants_nuclear_DE.csv`:

Site	Latitude	Longitude	SMR	MSR	LFR
Grohnde	51.906	9.401	✓	✓	✓
Brokdorf	53.851	9.345	✓	✓	✓

Site	Latitude	Longitude	SMR	MSR	LFR
Isar	48.617	12.293	✓	✓	—
Emsland	52.471	7.321	✓	—	✓
Neckarwestheim	49.040	9.175	✓	—	—

Each site is mapped to the nearest network bus by Euclidean distance in geographic coordinates. Generators are added with zero initial capacity ($p_{nom} = 0$) and $p_{nom_extendable} = \text{True}$.

4. Mathematical model

The model is a **linear programming (LP)** formulation: linear optimal power flow combined with capacity expansion. PyPSA constructs the optimisation problem; INRE modifies input time series and adds generators before the solve step.

4.1 Sets and indices

Symbol	Description
$\mathcal{N}$	Set of network buses (10 clustered zones)
$\mathcal{T}$	Set of snapshots (112 timesteps, 3-hour resolution)
$\mathcal{G}$	Set of generators
$\mathcal{L}$	Set of transmission lines
$\mathcal{S}$	Set of storage units and stores
$\mathcal{K}$	Set of energy carriers (solar, onwind, CCGT, nuclear-smr, ...)

4.2 Decision variables

Variable	Domain	Description
$p_{g,t}$	$\mathbb{R}_+$	Active power dispatch of generator g at time t (MW)
p_{nom_g}	$\mathbb{R}_+$	Installed capacity of extendable generator g (MW)
$f_{\ell,t}$	$\mathbb{R}$	Power flow on line ℓ at time t (MW)
$e_{s,t}$	$\mathbb{R}_+$	State of charge of storage s at time t (MWh)
$h_{s,t}$	$\mathbb{R}$	Charge/discharge power of storage s at time t (MW)

For non-extendable generators, p_{nom_g} is fixed to the installed capacity from the power plant database.

4.3 Parameters

Parameter	Description	Source
$d_{n,t}$	Electricity demand at bus n , time t	ENTSO-E
$\overline{p}_{g,t}$	Per-unit availability: $p_{max_pu}_{g,t}$	Atlite (modified by INRE for Dunkelflaute)
c_{cap_g}	Annuitised capital cost (EUR/MW/year)	technology-data + custom nuclear costs
c_{marg_g}	Marginal operating cost (EUR/MWh)	technology-data + custom nuclear costs
η_g	Electrical efficiency	technology-data / custom costs

Parameter	Description	Source
r_g	Ramp limit (p.u./hour)	Generator attributes
$\underline{p}_g$	Minimum stable load ($p_{\min_pu}$)	Generator attributes
e_{co2_g}	CO ₂ emissions intensity (t/MWh)	Carrier attributes
w_t	Snapshot weighting for objective (hours)	PyPSA snapshot weightings
$\overline{\ell}$	Line thermal capacity (MW)	PyPSA base network
$\overline{P}_g$	Maximum build limit ($p_{\text{nom_max}}$, MW)	Config / INRE script

Marginal cost for thermal generators combines variable O&M, fuel cost, and fixed O&M contribution:

$$\text{marginal_cost}_g = \frac{\text{VOM}_g + \text{fuel}_g \eta_g}{\eta_g} + \text{FOM contribution}_g$$

Annuited capital cost converts overnight investment to annual EUR/MW using the technology-data discount rate and component lifetime.

4.4 Objective function

The optimiser minimises **total system cost** over all snapshots, weighted by snapshot duration:

$$\min \sum_t w_t \left[\sum_g c_{\text{marg}_g} p_{g,t} + \sum_s c_{\text{marg}_s} |h_{s,t}| \right] + \sum_g c_{\text{cap}_g} p_{\text{nom}_g} + \sum_s c_{\text{cap}_s} e_{\text{nom}_s}$$

where G^{ext} and S^{ext} are sets of extendable generators and storage. The first term is operating expenditure (OPEX); the second and third terms are annuited capital expenditure (CAPEX) for new capacity.

Snapshot weightings ensure that costs integrated over the simulation window are scaled consistently with the configured planning period. For a 2-week window at 3-hour resolution, the simulated period represents approximately **3.8%** of a full year; investment decisions are therefore based on limited operational exposure (see Section 6).

4.5 Constraints

4.5.1 Nodal power balance (Kirchhoff's current law)

For each bus n and time t :

$$\sum_g p_{g,t} + \sum_{\ell: \text{to}(\ell)=n} f_{\ell,t} - \sum_{\ell: \text{from}(\ell)=n} f_{\ell,t} + \sum_s h_{s,t} = d_{n,t}$$

4.5.2 Generator dispatch limits

For each generator g and time t :

$$\underline{p}_g \cdot p_{\text{nom}_g} \leq p_{g,t} \leq \overline{p}_{g,t} \cdot p_{\text{nom}_g}$$

where $\overline{p}_{g,t}$ is the time-varying availability ($p_{\text{max_pu}}$), and $\underline{p}_g$ is the minimum stable load fraction ($p_{\text{min_pu}}$).

4.5.3 Ramp rate limits

For consecutive snapshots t and $t+1$:

$$p_{g,t+1} - p_{g,t} \leq r_g \cdot p_{\text{nom}_g} \quad p_{g,t} - p_{g,t+1} \leq r_g \cdot p_{\text{nom}_g}$$

4.5.4 Capacity expansion bounds

For extendable generators:

$$0 \leq p_{\text{nom},g} \leq \overline{P}_g$$

Nuclear generators receive $\overline{P}_g = 1,500$ MW per site from configuration.

4.5.5 Transmission line limits

$$|f_{\ell,t}| \leq \overline{f}_{\ell}$$

4.5.6 Storage dynamics

State-of-charge evolves with charge/discharge efficiency and standing losses (PyPSA standard storage formulation).

4.5.7 CO₂ emission cap

A global constraint limits total CO₂ emissions over the simulation period:

$$\sum_{t \in \mathcal{T}} w_t \sum_{g \in \mathcal{G}} e_{\text{co2},g} \cdot p_{g,t} \leq \text{CO2Limit} \cdot \frac{\sum_t w_t}{8760}$$

Configured annual cap: **500 Mt CO₂/year** (`electricity.co2limit: 500.e+6`). PyPSA scales this to the snapshot window duration.

4.5.8 Renewable resource limits

For VRE generators, dispatch is bounded by weather-derived availability profiles; no explicit energy budget constraint beyond the per-timestep $p_{\text{max},pu}$ limit.

5. INRE data processing and scenario modifications

PyPSA-Eur has no built-in Dunkelflaute or advanced nuclear logic. INRE applies modifications in `scripts/inre/apply_inre_network.py` before the LP solve.

5.1 Dunkelflaute stress model

Purpose: Represent extended periods of low wind and solar output characteristic of Central European winter anticyclones ("Dunkelflaute").

Implementation: `scripts/inre/apply_dunkelflaute.py` modifies `generators_t.p_max_pu` in place.

Parameters (`data/inre/dunkelflaute.yaml`):

Parameter	Value	Meaning
<code>wind_factor</code>	0.15	Retain 15% of normal wind availability during stress
<code>solar_factor</code>	0.10	Retain 10% of normal solar availability during stress
<code>auto_worst_days</code>	5	Auto-select 5 lowest-VRE days in the simulation window
<code>ramp_hours</code>	6	Smooth transition over 6 snapshots (18 h) at window edges
<code>time_start / time_end</code>	2021-01-28 → 2021-02-03	Fallback fixed window (used when <code>auto_worst_days</code> is null)

Stress day selection (auto mode):

For each day d in the simulation window, compute a capacity-weighted VRE score:

$$\text{score}_d = \frac{1}{T_d} \sum_{t \in T_d} \left[\sum_{g \in \text{wind}} p_{\max_pu_g,t} \cdot p_{\text{nom}_g} + \sum_{g \in \text{solar}} p_{\max_pu_g,t} \cdot p_{\text{nom}_g} \right]$$

The N days with the lowest score are selected as the stress window. Lower score indicates worse Dunkelflaute conditions.

Availability derating:

Inside the stress mask, availability is scaled:

$$p_{\max_pu_g,t} \rightarrow p_{\max_pu_g,t} \cdot f_{\{\text{carrier}\}(g)}$$

where $f = 0.15$ for wind carriers (`onwind`, `offwind-ac`, `offwind-dc`, `offwind-float`) and $f = 0.10$ for solar carriers (`solar`, `solar-hsat`).

Edge ramping: At the boundaries of the stress window, a linear ramp over `ramp_hours` snapshots blends between full and stressed availability to avoid discontinuous step changes.

Spatial treatment: Derating is applied **uniformly across all nodes** (national correlation assumption).

5.2 Advanced nuclear technology addition

Implementation: `scripts/inre/add_nuclear_technologies.py`

For each row in the sites CSV matching the scenario carrier:

1. Find the nearest network bus by geographic distance.
2. Add an extendable `Generator` component with:
 - `p_nom = 0`, `p_nom_extendable = True`
 - `p_nom_max` from config (default 1,500 MW)
 - `capital_cost` and `marginal_cost` from processed 2050 costs
 - `p_max_pu = 0.9`, `p_min_pu = 0.3`, `ramp_limit = 0.5`
3. Register the carrier in the network if not already present.

The LP optimiser then decides optimal installed capacity p_{nom_g} balancing annuitised CAPEX against dispatch value over the snapshot window.

5.3 Cost data integration

Custom nuclear costs in `data/inre/custom_costs_nuclear.csv` follow the PyPSA technology-data format:

planning_horizon, technology, parameter, value, unit, source, further description

During the build phase, this file is merged via `costs.custom_cost_fn`. The processed output (`costs_2050_processed.csv`) provides `capital_cost` (EUR/MW/year) and `marginal_cost` (EUR/MWh) used when adding nuclear generators.

Investment values in the CSV include decommissioning:

$$\text{investment}_{\text{total}} = \text{CAPEX} \cdot 1.15$$

5.4 Scenario matrix

Five scenarios are defined in `config/inre/scenarios.yaml`:

Scenario	Dunkelflaute	New nuclear carrier	Purpose
<code>base</code>	Off	None	Reference winter dispatch
<code>dunkelflaute</code>	On	None	Quantify adequacy gap under stress
<code>dunkelflaute-smr</code>	On	<code>nuclear-smr</code> (5 sites)	SMR contribution under stress
<code>dunkelflaute-msr</code>	On	<code>nuclear-msr</code> (3 sites)	MSR contribution under stress
<code>dunkelflaute-lfr</code>	On	<code>nuclear-lfr</code> (3 sites)	LFR contribution under stress

Shared build resources (`run.shared_resources.policy: true`) mean grid topology, clustering, weather profiles, and the prepared network are built once. Only the INRE modification step and LP solve differ per scenario.

6. Solution method and output metrics

6.1 Solver

The LP is solved with **HIGHS** (interior-point method). Typical solve time: 5–7 minutes per scenario (10 clusters, 112 snapshots).

6.2 Key performance indicators

KPI	Calculation
Operating cost (OPEX)	<code>n.statistics.opex()</code> — variable + fixed O&M over weighted snapshots
Investment (CAPEX)	<code>n.statistics.capex()</code> — annuitised capital for optimal capacity
Total objective	<code>n.objective</code> — LP objective value
Generation by carrier	<code>n.generators_t.p</code> × snapshot weights, summed
Optimal capacity	<code>n.generators.p_nom_opt</code> for extendable units
CO ₂ emissions	$\sum_{g,t} p_{g,t} \cdot w_t \cdot e_{co2,g}$

7. Assumptions and limitations

7.1 Key assumptions

ID	Assumption	Value	Sensitivity
T1	Modelled country	Germany only	High — no imports
T2	Spatial resolution	10 clusters	Medium
T3	Simulation window	2 weeks (Jan 2021)	High — limits investment signal
T4	Temporal resolution	3 hours	Medium
T5	Cost year	2050	High
D1	Wind derating factor	0.15	Very high
D2	Solar derating factor	0.10	Very high
D3	Stress duration	5 auto-selected days	High

ID	Assumption	Value	Sensitivity
P1	CO ₂ cap	500 Mt/year	High — non-binding in current results
P2	Existing nuclear	0 MW	Correct for post-2023 Germany

7.2 Known limitations

- 1. **Short snapshot window:** Investment decisions are based on ~3.8% of a year; nuclear and storage economics are not representative of full annual or lifetime utilisation.
- 2. **Linear programming:** No unit commitment, start-up costs, or integer build decisions.
- 3. **National Dunkelflaute:** Uniform derating ignores spatial variation in cloud cover and wind patterns.
- 4. **No cross-border flows:** Germany-only; neighbouring countries not modelled.
- 5. **CO₂ cap:** Current 500 Mt/year limit is non-binding; coal and lignite remain in the dispatch stack.
- 6. **Placeholder elements:** Fuel costs for nuclear and some operational parameters are INRE assumptions pending full literature review.

8. Reproducibility

8.1 Run commands

```
# Environment
pixi install && pixi shell

# All five scenarios
snakemake -call solve_elec_networks \
  --configfile config/inre/config.base.yaml \
  --configfile config/inre/config.scenarios.yaml

# Cross-scenario comparison
python scripts/inre/compare_scenarios.py --output-dir results/inre-comparison
```

8.2 Key configuration files

File	Role
config/inre/config.base.yaml	Base case parameters
config/inre/config.scenarios.yaml	Multi-scenario driver
config/inre/scenarios.yaml	Per-scenario overrides
data/inre/dunkelflaute.yaml	Dunkelflaute stress parameters
data/inre/custom_costs_nuclear.csv	Nuclear techno-economic data
data/inre/custom_powerplants_nuclear_DE.csv	Candidate reactor sites

8.3 INRE scripts

Script	Function
scripts/inre/apply_inre_network.py	Orchestrator (Snakemake entry point)
scripts/inre/apply_dunkelflaute.py	VRE profile derating

Script	Function
<code>scripts/inre/add_nuclear_technologies.py</code>	Extendable nuclear generator addition
<code>scripts/inre/compare_scenarios.py</code>	Cross-scenario KPI comparison

9. References

- PyPSA documentation: [Linear optimal power flow](#)
- PyPSA-Eur: <https://pypsa-eur.readthedocs.io/>
- PyPSA technology-data: <https://github.com/PyPSA/technology-data>
- OECD/NEA, IEA: *Projected Costs of Generating Electricity* (nuclear cost basis)
- INRE project documentation: `INRE-README.md`, `INRE-PROGRESS-REPORT.md`

Document history

Version	Date	Notes
1.0	2026-07-01	Initial methodology document for report preparation